



UNIVERSIDAD

FHEC

FEDERICO HENRÍQUEZ Y CARVAJAL

Materia:

Base de Datos II

Maestro:

Ing. Jaime Vilorio Green

Autor:

Brayan Napoleon,

Matriculas:

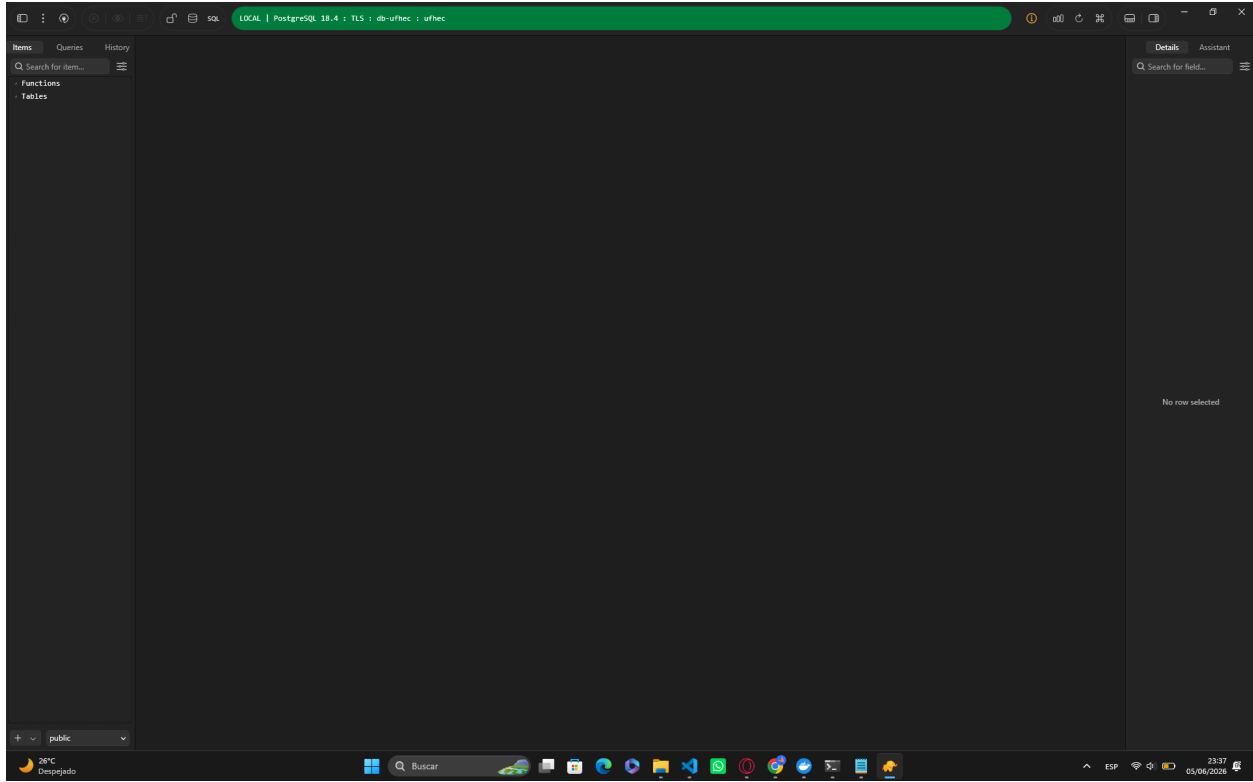
LR-2024-02332

Tema:

Gestión de Transacciones

Fecha:

06/06/2026



cuenta				SQL Query		usuarios		SQL Query	
id	nombre	apellido	email						
1	Alex	Pena	alex@ufhec.edu.do						
3	Jose	Pewef	DMF5o@ufhec.edu.do						

LOCAL | Executed Statements Count: 0/1

Items Queries History SQL Query

Search for item...

Functions

Tables

► cuenta

► usuarios

```
1 BEGIN;
2
3 UPDATE cuenta
4 SET monto = monto + 50
5 WHERE usuario_id = 1;
6
7 SELECT pg_backend_pid();
8
9 SELECT pid,
10        username,
11        state,
12        query
13 FROM pg_stat_activity
14 WHERE state <> 'idle';
15
16 COMMIT;
17
18 SELECT u.username, c.monto
19 FROM usuarios u
20 JOIN cuenta c ON u.id = c.usuario_id;
21
22
23 CREATE OR REPLACE FUNCTION transferir(
24     origen INTEGER,
25     destino INTEGER,
26     cantidad NUMERIC
27 )
28 RETURNS VOID AS
29 $$
30 DECLARE
31     saldo NUMERIC;
32 BEGIN
33     SELECT monto
34     INTO saldo
35     FROM cuenta
36     WHERE usuario_id = origen
37     FOR UPDATE;
38     IF saldo < cantidad THEN
39         RAISE EXCEPTION 'Fondos insuficientes';
40     END IF;
41     UPDATE cuenta
42     SET monto = monto - cantidad
43     WHERE usuario_id = origen;
44     UPDATE cuenta
45     SET monto = monto + cantidad
46     WHERE usuario_id = destino;
47 END;
48
49 $$ LANGUAGE plpgsql;
50
51 BEGIN;
52 SELECT transferir(2, 1, 5000);
53
54
```

line 17, column 1, location 219

No limit Beautify Cancel

Details Assistant

Search for field...

No row selected

public Data Message Chart Waiting 0.0 s 0 rows Export...

26°C Mayorm, sábado

Windows taskbar with various icons and system clock showing 23:10 on 06/04/2026.

LOCAL | PostgreSQL 18.4 : TLS : db-ufhec : ufhec : SQL Query

Items Queries History SQL Query

Search for item...

Functions

Tables

► cuenta

► usuarios

```
22
23 CREATE OR REPLACE FUNCTION transferir(
24     origen INTEGER,
25     destino INTEGER,
26     cantidad NUMERIC
27 )
28 RETURNS VOID AS
29 $$
30 DECLARE
31     saldo NUMERIC;
32 BEGIN
33     SELECT monto
34     INTO saldo
35     FROM cuenta
36     WHERE usuario_id = origen
37     FOR UPDATE;
38     IF saldo < cantidad THEN
39         RAISE EXCEPTION 'Fondos insuficientes';
40     END IF;
41     UPDATE cuenta
42     SET monto = monto - cantidad
43     WHERE usuario_id = origen;
44     UPDATE cuenta
45     SET monto = monto + cantidad
46     WHERE usuario_id = destino;
47 END;
48
49 $$ LANGUAGE plpgsql;
50
51 BEGIN;
52 SELECT transferir(2, 1, 5000);
53
54
55
```

line 61, column 1, location 922

Query 1 OK: ROLLBACK

No limit Beautify Run Current

Details Assistant

Search for field...

No row selected

Windows taskbar with various icons and system clock showing 23:10 on 06/04/2026.

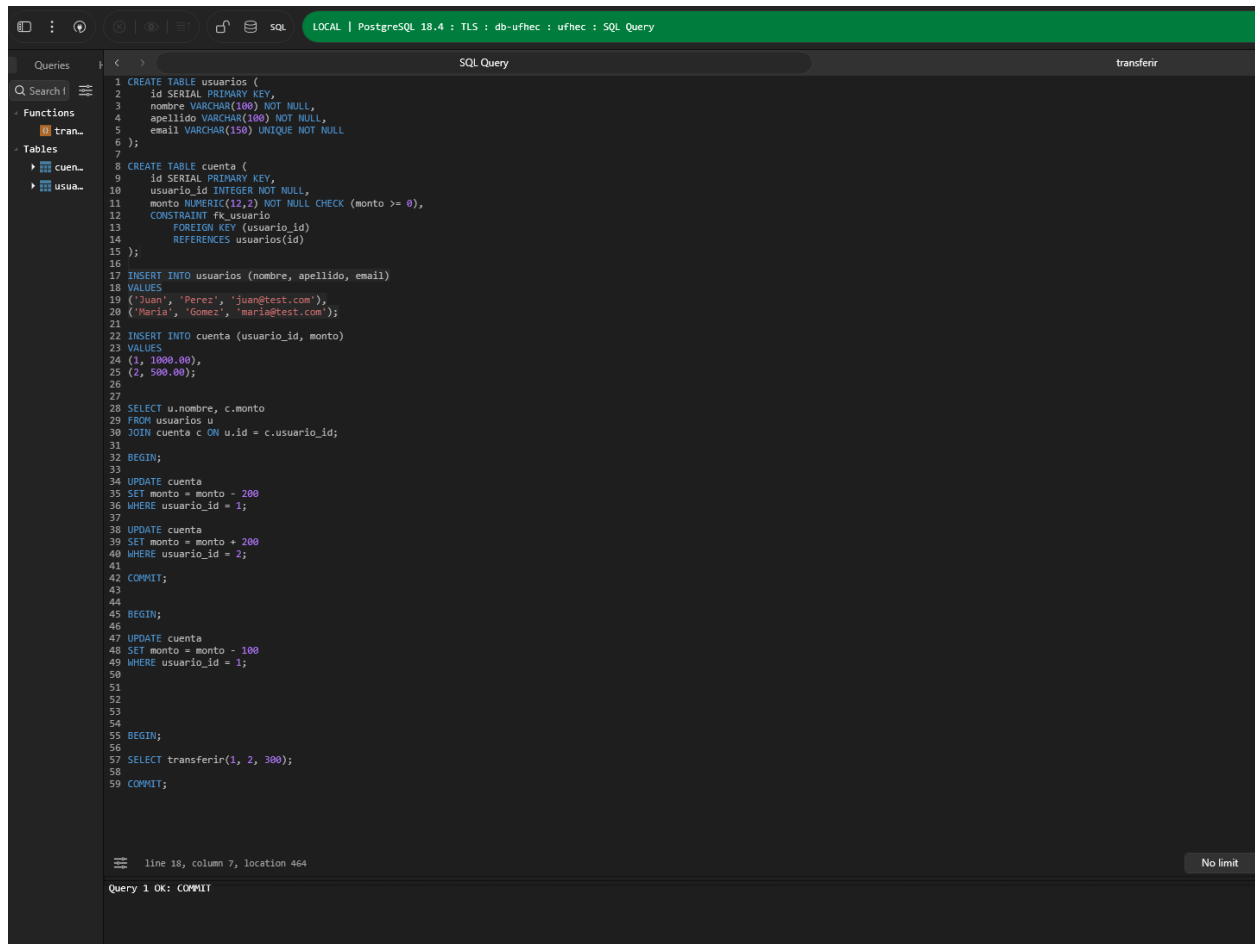
The screenshot displays a PostgreSQL SQL Editor interface with a dark theme. The main window shows a SQL script with the following content:

```
1 BEGIN;
2
3 UPDATE cuenta
4 SET monto = monto + 50
5 WHERE usuario_id = 1;
6
7 SELECT pg_backend_pid();
8
9 SELECT pid,
10 username,
11 state,
12 query
13 FROM pg_stat_activity
14 WHERE state <> 'idle';
15
16 COMMIT;
17
18 SELECT u.nombre, c.monto
19 FROM usuarios u
20 JOIN cuenta c ON u.id = c.usuario_id;
21
22 CREATE OR REPLACE FUNCTION transferir(
23     origen INTEGER,
24     destino INTEGER,
25     cantidad NUMERIC
26 )
27 RETURNS VOID AS
28 $$
29 BEGIN
30     saldo NUMERIC;
31
32 BEGIN
33     SELECT monto
34     INTO saldo
35     FROM cuenta
36     WHERE usuario_id = origen
37     FOR UPDATE;
38
39 IF saldo < cantidad THEN
40     RAISE EXCEPTION 'Saldo insuficiente';
41 END IF;
42
43 UPDATE cuenta
44 SET monto = monto - cantidad
45 WHERE usuario_id = origen
46 FOR UPDATE;
47
48 UPDATE cuenta
49 SET monto = monto + cantidad
50 WHERE usuario_id = destino
51 FOR UPDATE;
52
53 COMMIT;
54 END;
```

The script is executed, and the results are displayed in a table with the following data:

nombre	monto
Juan	250.00
Maria	1300.00

The interface also shows a sidebar with a search bar and a list of functions and tables. The bottom status bar indicates the current database is 'public' and the connection is 'Waiting 18.6 s'.

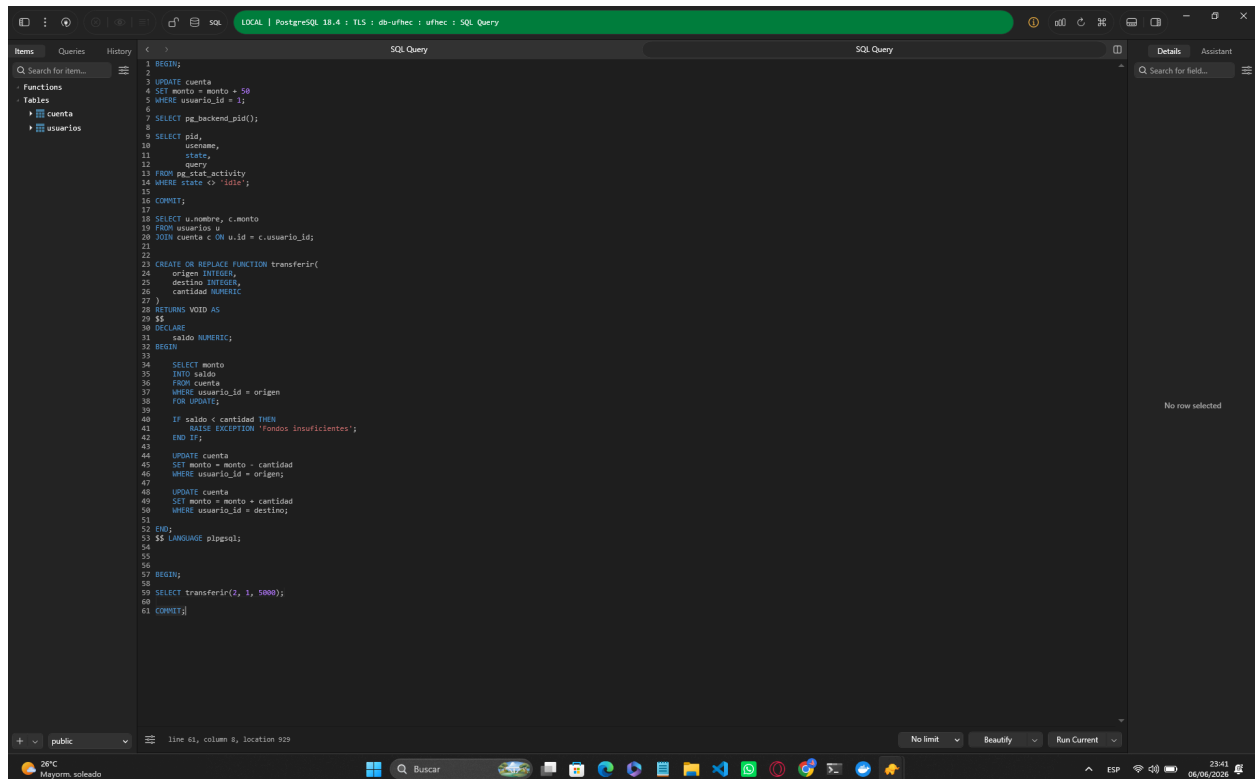


The screenshot shows a PostgreSQL SQL query editor interface. The title bar indicates the connection is 'LOCAL | PostgreSQL 18.4 : TLS : db-ufhec : ufhec : SQL Query'. The left sidebar contains a 'Queries' panel with a search bar and a 'Tables' section showing 'cuen...' and 'usua...'. The main editor area displays an SQL script with line numbers 1 through 59. The script includes the following commands:

```
1 CREATE TABLE usuarios (  
2   id SERIAL PRIMARY KEY,  
3   nombre VARCHAR(100) NOT NULL,  
4   apellido VARCHAR(100) NOT NULL,  
5   email VARCHAR(150) UNIQUE NOT NULL  
6 );  
7  
8 CREATE TABLE cuenta (  
9   id SERIAL PRIMARY KEY,  
10  usuario_id INTEGER NOT NULL,  
11  monto NUMERIC(12,2) NOT NULL CHECK (monto >= 0),  
12  CONSTRAINT fk_usuario  
13    FOREIGN KEY (usuario_id)  
14    REFERENCES usuarios(id)  
15 );  
16  
17 INSERT INTO usuarios (nombre, apellido, email)  
18 VALUES  
19 ('Juan', 'Perez', 'juan@test.com'),  
20 ('Maria', 'Gomez', 'maria@test.com');  
21  
22 INSERT INTO cuenta (usuario_id, monto)  
23 VALUES  
24 (1, 1000.00),  
25 (2, 500.00);  
26  
27  
28 SELECT u.nombre, c.monto  
29 FROM usuarios u  
30 JOIN cuenta c ON u.id = c.usuario_id;  
31  
32 BEGIN;  
33  
34 UPDATE cuenta  
35 SET monto = monto - 200  
36 WHERE usuario_id = 1;  
37  
38 UPDATE cuenta  
39 SET monto = monto + 200  
40 WHERE usuario_id = 2;  
41  
42 COMMIT;  
43  
44  
45 BEGIN;  
46  
47 UPDATE cuenta  
48 SET monto = monto - 100  
49 WHERE usuario_id = 1;  
50  
51  
52  
53  
54  
55 BEGIN;  
56  
57 SELECT transferir(1, 2, 300);  
58  
59 COMMIT;
```

At the bottom of the editor, a status bar shows 'line 18, column 7, location 464' and a message 'Query 1 OK: COMMIT'. A 'No limit' button is visible in the bottom right corner.

la primera sesion



segunda sesion